

ONLINE SHOPPING SYSTEM

Assignment report

submitted by

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Introduction

The Software Requirements Specification (SRS) document provides a detailed description of the Online Shopping System. This system is designed to facilitate, manage, and streamline online retail operations in real time. It includes features such as product browsing, order management, payment processing, and reporting. The document serves as a guideline for developers, testers, and stakeholders to understand system functionality and constraints.

1.1 Purpose:

The purpose of this document is to clearly define the requirements of the Online Shopping System. It helps:

- Developers understand system functionalities and implementation needs
- Stakeholders review system objectives and expected outcomes
- Testers verify system performance against requirements
- Administrators manage and monitor shopping operations efficiently

1.2 Product Scope:

The Online Shopping System is designed to manage and streamline e-commerce operations efficiently. The key features include:

- Real-time monitoring of orders, inventory, and sales performance
- Detection and alerting of payment failures and stock shortages
- Visualization of sales statistics through dashboards
- Data logging and report generation for analysis
- Secure user authentication and role-based access

The system aims to improve customer experience, reduce cart abandonment, and ensure optimal performance of online shopping services.

1.3 Definitions, Acronyms, and Abbreviations

- OSS: Online Shopping System

- SRS: Software Requirements Specification
- UI: User Interface
- KPI: Key Performance Indicator
- QoS: Quality of Service
- API: Application Programming Interface
- Admin: System administrator with full access
- User: Authorized person who accesses the shopping platform

1.4 Technologies to be used

The following technologies will be used for developing the Online Shopping System:

- Frontend: HTML, CSS, JavaScript (React.js optional)
- Backend: Python (Django/Flask) or Node.js
- Database: MySQL / PostgreSQL / MongoDB
- Data Processing & Analytics: Python libraries (Pandas, NumPy)
- Monitoring Tools: Prometheus, Grafana (for visualization)
- APIs: REST APIs for communication
- Deployment Tools: Docker, Kubernetes (optional)
- Operating System: Linux / Windows

Overall Description

The Online Shopping System is designed to manage and streamline e-commerce operations in real time. It collects data from order and inventory components and displays key metrics such as sales volume, order status, and customer activity through a user-friendly dashboard. The system helps detect failures, generate alerts, and provide reports for better decision-making. It ensures efficient platform performance, reduced downtime, and secure access for users.

2.1 Product Perspective

The Online Shopping System is a standalone system that integrates with existing e-commerce infrastructure. It interacts with payment gateways, inventory systems, and external tools to collect and analyse performance data, providing a centralized platform for monitoring and management.

2.2 User Characteristics and Classes

User Characteristics

- Users include both technical (developers, admins) and non-technical (customers, managers).
- Varying experience levels: expert, intermediate, novice.
- Users have role-based access (admin, seller, customer).
- Admins use the system continuously, while customers use it for browsing and purchasing.

2.3 Classes of Users

- System Administrator – Full control, system configuration and user management.
- Seller/Vendor – Manages products, inventory, and order fulfillment.
- Customer – Browses products, places orders, and tracks deliveries.
- Manager/Analyst – Views reports and sales metrics.

- Support Staff – Handles customer queries and order issues.
- Security Analyst – Monitors security and prevents fraudulent transactions.

The Online Shopping System operates on cloud and on-premise servers with support for Linux and Windows operating systems. It requires a high-speed internet connection and is accessible through web-based interfaces (browsers). The system integrates with payment gateways, databases, and analytics tools, and supports real-time data processing and reporting.

2.3.2 User Documentation

The system provides user manuals, installation guides, and online help documentation to assist users in operating and managing the Online Shopping System. It includes step-by-step instructions, troubleshooting guides, and FAQs for different user roles.

2.3.3 Software Interfaces

The Online Shopping System interfaces with payment gateways, inventory management systems, and external logistics tools for order processing. It supports integration through APIs, protocols, and web services to ensure seamless communication between system components.

2.3.4 Hardware Interfaces

The system interacts with servers, barcode scanners, payment terminals, and networking devices for collecting and processing real-time transaction data.

2.3.5 External Forces

The system is affected by network traffic load, payment gateway availability, supply chain disruptions, and regulatory policies, which may impact performance.

2.3.6 Dependencies

The system depends on stable network connectivity, cloud services, payment APIs, and third-party logistics services for proper functioning.

2.3.1. Operating Environment

2.3.7 Constraints

The system is constrained by hardware limitations, bandwidth availability, payment compliance (PCI-DSS), and adherence to e-commerce regulations.

3.3 System Features

3.3.1 Functional Requirements

- Monitor real-time order processing (status, quantity, delivery time).
- Detect and report payment failures or inventory shortages.
- Generate alerts and notifications for abnormal conditions.
- Provide dashboards and reports for analysis.
- Allow user management and role-based access control.

3.3.2 Non-Functional Requirements

- **Performance:** System should support real-time data processing.
- **Reliability:** Ensure high availability with minimal downtime.
- **Scalability:** Handle large-scale e-commerce transaction data.
- **Security:** Protect data with authentication and encryption.
- **Usability:** Provide a user-friendly interface.

Commitment to delivering a high-quality university website

The system is committed to delivering a reliable, secure, and user-friendly university website that ensures seamless access to information and services. It focuses on high performance, accessibility, data security, and regular updates to meet user needs and maintain quality standards.

How testing contributes to user satisfaction and operational efficiency

Testing ensures that the system is reliable, error-free, and performs as expected, leading to improved user satisfaction. It helps identify and fix issues early, ensuring a smooth and user-friendly experience. Additionally, testing enhances operational efficiency by reducing system failures, improving performance, and minimizing maintenance costs.

ONLINE SHOPPING SYSTEM DESIGN DOCUMENT

5.1 System Design (front-end design)

5.1.1 Use case diagram

The use case diagram illustrates the interactions between system actors (Customer, Seller, Admin, Payment Gateway) and system functions such as browsing products, placing orders, managing inventory, processing payments, and generating reports.

Actor	Use Cases
Customer	Browse Products, Add to Cart, Place Order, Track Order, Make Payment
Seller/Admin	Manage Products, View Orders, Update Inventory, Generate Reports
Payment Gateway	Process Payment, Verify Transaction, Send Confirmation
System Admin	Manage Users, Configure System, Monitor Performance

5.1.2 Class Diagram

The class diagram defines the structure of the Online Shopping System, including classes such as User, Product, Order, Cart, Payment, and their relationships. Key classes include:

- **User** – ID, Username, Role, Email, Login(), Logout()
- **Product** – ProductID, Name, Price, Stock, addProduct(), updateStock()
- **Order** – OrderID, Status, Timestamp, placeOrder(), cancelOrder()
- **Cart** – CartID, Items, addItem(), removeItem(), checkout()
- **Payment** – PaymentID, Method, Status, processPayment(), refund()

5.1.3 Sequence Diagram

The sequence diagram for the Online Shopping System shows the step-by-step interaction flow:

1. Customer logs in through the Web UI
2. UI sends authentication request to Backend Server
3. Backend verifies credentials and returns Auth Success
4. Dashboard is displayed to the Customer
5. Customer browses products and adds items to cart
6. UI requests product data from Backend
7. Backend fetches product details from Database
8. Customer proceeds to checkout and initiates payment
9. Backend communicates with Payment Gateway
10. Payment Gateway processes transaction and returns status
11. Order is confirmed and stored in Database
12. Confirmation notification sent to Customer
13. Seller receives order notification
14. Admin monitors order analytics on dashboard

5.1.4 Collaboration Diagram

The collaboration diagram for the Online Shopping System illustrates the object interactions:

Interaction Flow Summary:

1. Customer logs in through Web UI
- 2-3. UI authenticates with Backend
4. UI shows product dashboard
- 5-6. UI requests product data from Backend
7. Backend fetches data from Database
- 8-9. Product data displayed to Customer
10. Customer adds to cart and initiates checkout
11. Backend processes order and payment
12. Database stores order record
13. UI shows order confirmation to Customer
- 14-15. Customer sets delivery preferences
- 16-17. Backend triggers fulfillment process
- 18-19. Backend notifies UI; UI notifies Customer

5.1.5 Data Flow Diagram

Level 0 DFD

The Level 0 DFD shows the Online Shopping System as a single process receiving inputs from Customers (login, product search, orders), Sellers (product listings, inventory updates), and Payment Gateways (transaction confirmations). Outputs include order confirmations, invoices, dashboards, and notifications stored in the central Database.

External Entity	Input to System	Output from System
Customer / Admin	Login, Search, Orders, Payments	Dashboard, Confirmations, Alerts
Payment Gateway	Transaction Status	Payment Requests
Database	Stored Data	Store / Retrieve Data

Level 1 DFD

The Level 1 DFD decomposes the system into four main processes:

- **1.1 Data Collection** – Receives raw data from product listings, customer actions, and payment systems → stores in Network Data Logs (D1)
- **1.2 Data Processing & Analysis** – Processes collected data; aggregates order metrics and sales KPIs
- **1.3 Alert Generation** – Produces alerts for failed payments, low stock, or unusual activity → stored in Alerts Database (D2)
- **1.4 Report Management** – Generates sales and performance reports → stored in Reports Database (D3)

The User/Admin receives Alerts and Reports from D2 and D3 respectively.

5.2 Database Design

5.2.1 ER Diagram

The ER Diagram for the Online Shopping System includes the following entities and relationships:

Entity	Attributes	Relationships
USER/ADMIN	AdminID (PK), Name, Role, Email, AccessLevel	GENERATES Reports (1:N)
PRODUCT	ProductID (PK), Name, Category, Price, Stock, Status	BELONGS TO Order (N:M)
ORDER	OrderID (PK), CustomerID, Status, DateCreated, TotalAmount	PROCESSED FROM Cart (N:1)
CART	CartID (PK), CustomerID, Items, CreatedAt	GENERATES Order (1:N)
PAYMENT	PaymentID (PK), Method, Status, Timestamp, Amount	TRIGGERS Alert (1:N)
ALERT	AlertID (PK), Severity, Message, Timestamp, Status	SENDS TO Admin (N:1)
REPORT	ReportID (PK), Type, DateCreated, Content (Sales Metrics)	GENERATED BY Admin (N:1)

Future scope

The Online Shopping System has strong potential for expansion and enhancement as technology evolves. The following points outline its future scope:

6.1. Integration with AI & Machine Learning

- Implement intelligent algorithms for **predictive analysis** of customer behaviour
- Enable **personalised recommendations** by automatically detecting user preferences
- Improve fraud detection using real-time learning models

6.2 Support for Advanced Payment Systems

- Integrate with cryptocurrency and **next-generation payment technologies**
- Handle higher transaction volumes, ultra-low latency, and advanced security protocols

6.3. Real-Time Advanced Analytics

- Incorporate **big data analytics** for handling massive sales traffic
- Provide deeper insights through dashboards with real-time visualization

6.4. IoT Device Monitoring Expansion

- Extend monitoring to smart retail devices and **IoT sensors**
- Ensure efficient management of smart warehouses, healthcare retail, and industrial supply chains
- Ensure efficient management of smart cities, healthcare, and industrial IoT systems
- 5. Cloud & Edge Computing Inte
- Use **edge computing** for faster local data processing and reduced latency

6.5. Cloud & Edge Computing Integration

- Deploy system on **cloud platforms** for scalability and flexibility
- Use **edge computing** for faster local data processing and reduced latency

6.6. Enhanced Security Features

- Implement advanced cybersecurity mechanisms to prevent attacks
- Include intrusion detection and prevention systems (IDPS)
- Ensure secure data transmission and storage

6.7. Automation & Self-Optimization

- Enable automatic **inventory optimization** based on sales patterns
- Reduce human intervention through automated decision-making systems
- Connect with **order management systems (OMS)** and enterprise tools
- Enable seamless data exchange with third-party platforms

6.8. User Experience Enhancements

- Improve UI/UX with intuitive dashboards
- Provide customizable alerts, reports, and analytics views

Conclusion

The Software Requirements Specification (SRS) for the Online Shopping System defines the overall functionality, performance, and constraints of the system in a clear and structured manner. It outlines the key features such as real-time order monitoring, analysis, alert generation, and report management, ensuring efficient supervision of e-commerce operations.

This document serves as a foundation for developers, designers, and stakeholders to understand system requirements and expectations. By following this SRS, the system can be developed to be reliable, scalable, and secure, meeting the demands of modern high-speed digital commerce.

In conclusion, the Online Shopping System aims to enhance platform performance, reduce downtime, and provide accurate insights, making it an essential tool for managing next-generation e-commerce infrastructure.

Online Shopping System Testing Document

7.1. Introduction

7.1.1 Purpose

The purpose of this document is to outline the testing strategy and procedures for the Online Shopping System.

7.1.2 Scope

This document covers functional, performance, security, and usability testing of the Online Shopping System.

7.1.3 Objectives

- Ensure the reliability and correctness of the system
- Validate that the system meets specified requirements
- Identify and fix defects in the system

7.2. Test Plan

7.2.1 Test Strategy

The testing approach includes:

- Functional Testing
- Performance Testing
- Security Testing
- Usability Testing

It also defines test levels, entry/exit criteria, and testing techniques.

7.2.2 Test Environment

- Hardware: Servers, monitoring devices, user systems
- Software: OS (Windows/Linux), database, e-commerce tools
- Network: High-speed internet or simulated e-commerce environment

7.2.3 Test Schedule

Defines timeline for:

- Test planning
- Test case design
- Test execution
- Defect fixing & retesting

7.3 Functional Testing

7.3.1 Test Cases

Test cases will cover:

- Product browsing and search
- Data collection from orders/transactions
- Real-time order processing
- Alert generation for payment failures
- Report generation

- User/Admin login & management

7.3.2 Test Scenarios

- Monitoring real-time order status
- Detecting payment failures
- Generating alerts on low stock threshold breach
- Viewing reports and analytics

7.3.3 Test Data

- Valid and invalid product/order inputs
- Simulated transaction data
- Fault/fraud simulation data

7.3.4 Test Execution

Includes:

- Execution of test cases
- Defect reporting
- Retesting after fixes

7.4. Performance Testing

7.4.1 Load Testing

Test system performance under normal and peak traffic conditions.

7.4.2 Stress Testing

Evaluate system behaviour under extreme transaction load and failures.

7.4.3 Performance Metrics

- Response time
- Throughput
- Latency
- Resource utilization

7.5. Security Testing

7.5.1 Authentication and Authorization

Verify secure login and role-based access control.

7.5.2 Data Encryption

Ensure secure transmission and storage of payment and customer data.

7.5.3 Access Control

Prevent unauthorized access to sensitive shopping and transaction data.

7.6. Usability Testing

7.6.1 User Interface

Evaluate ease of use, dashboard clarity, and navigation.

7.6.2 Workflow Testing

Check efficiency of shopping workflows and order handling.

7.6.3 Accessibility

Ensure system is usable for all users, including accessibility considerations.

7.7. Defect Management

7.7.1 Defect Reporting

Track defects with severity and priority.

7.7.2 Regression Testing

Ensure new updates do not introduce new issues.

7.8. Conclusion

This document summarizes the testing strategy for the Online Shopping System. It ensures that the system is reliable, secure, and performs efficiently under various conditions. The document should be updated as the system evolves and new features are introduced.

7.9. References

1. Software Engineering textbooks
2. IEEE Testing Standards
3. Online tools (SmartDraw, Visual Paradigm)
4. Official documentation and resources